

Athary E-Learning Platform - Use Cases Document v2.0

Project: Athary Educational Platform

Version: 2.0

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Part 1: Introduction

Document Purpose

This document specifies all use cases for the Athary E-Learning Platform, following standardized format for consistency.

Actors

Actor	Description
Student	Enrolls in courses, learns, takes quizzes
Teacher	Creates courses, teaches, grades
Assistant	Helps teacher with grading and monitoring
Admin	Manages system, users, and platform
System	Automated processes and background tasks

Use Case Template Structure

Each UC includes: ID, Name, Actors, Goal, Scope, Trigger, Preconditions, Main Flow, Alternative Flows, Exception Flows, Postconditions, Requirements, Busi-

ness Rules, Notes.

Part 2: Use Cases

1. User Management

UC-001: Register New User

Primary Actor: New User

Goal: Create account on platform

Main Flow

1. User clicks “Create Account”
2. System shows registration options (Email or OAuth)
3. **Email Path:** User enters name, email, password → System validates → Creates account (Inactive) → Sends activation email → User activates → Account Active
4. **OAuth Path:** User clicks Google/Facebook/Apple → Authenticates with provider → System creates account (Active) → Auto-login

Key Features

- Email registration with activation
- OAuth 2.0 (Google, Facebook, Apple)
- Password strength validation
- reCAPTCHA verification
- Account linking capability

Business Rules

- Password: 8+ chars, mixed case, number, symbol
 - Email must be unique
 - OAuth accounts pre-verified (no activation needed)
 - Rate limit: 5 attempts/hour per IP
-

UC-002: Login

Primary Actor: Registered User

Goal: Authenticate and access platform

Main Flow

1. User opens login page
2. System shows login options
3. **Email Path:** Enters email/password → System validates → Creates JWT session → Redirects to dashboard
4. **OAuth Path:** Clicks provider → Authenticates → System finds/creates session → Redirects

Key Features

- Email/password authentication
- Social login (Google, Facebook, Apple)
- Remember Me (30 days)
- 2FA support
- Account linking during login

Business Rules

- 5 failed attempts = 30min logout
 - Session: 1h standard, 30 days with Remember Me
 - Auto-logout after 30min inactivity
 - Log all login attempts
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UC-003: Password Recovery

Primary Actor: Registered User

Goal: Reset forgotten password

Main Flow

1. User clicks “Forgot Password?”
2. Enters email → System generates reset token (1h validity)
3. System sends reset email with link
4. User clicks link → Enters new password → System validates
5. System updates password → Invalidates token → Sends confirmation

Business Rules

- Reset token: 1h validity, single-use
 - Rate limit: 3 resets/hour per email
 - Cannot reuse last 3 passwords
 - Force logout all devices after change
-

UC-004: Update Profile

Primary Actor: Any Logged-in User

Goal: Update personal information

Main Flow

1. User opens Profile Settings
2. System displays edit form
3. User modifies fields (name, picture, phone, bio)
4. System validates → Updates database → Clears cache

Editable Fields

- Full Name, Profile Picture, Phone Number
- Bio (teachers), Social Links
- Notification Preferences, Language, Timezone

Business Rules

- Picture: Max 2MB, JPG/PNG/WebP, min 200x200px
 - Name: 3-50 chars
 - Email change: Separate process, max once/30 days
-

UC-005: Manage Connected Accounts (OAuth)

Primary Actor: Registered User

Goal: Manage linked social login accounts

Main Flow

1. User opens Connected Accounts settings
2. System shows all linked methods (Email, Google, Facebook, Apple)
3. User can:
 - Link new provider
 - Unlink existing provider (if >1 method exists)
 - Set primary method

Business Rules

- Must have 1 active authentication method
 - Max 3 OAuth providers per account
 - Cannot unlink last method
 - Unlinking requires password confirmation
-

2. Course Management

UC-006: Create Course

Primary Actor: Teacher

Goal: Create new course on platform

Main Flow (5-Step Wizard)

1. **Step 1:** Basic info (title, description, category, level, language)
2. **Step 2:** Pricing (free/paid, price, teaching method, schedule)
3. **Step 3:** Media (cover image, preview video)
4. **Step 4:** Course structure (sections, estimated content)
5. **Step 5:** Review all → Create course (status: Draft)

Business Rules

- Title: 10-100 chars, unique per teacher
 - Price: Free (0) or Paid (10-10,000 SAR)
 - Cover image required before publishing
 - Free teachers: max 3 courses
 - Verified teachers: unlimited
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UC-007: Edit Course

Primary Actor: Teacher

Goal: Modify course information

Main Flow

1. Teacher selects course → Clicks “Edit”
2. System loads current data → Shows edit form
3. Teacher modifies fields → Saves changes
4. If published course: System notifies enrolled students

Editable Fields

- Always: Description, tags, promotional content
- With restrictions: Title (no enrollments), Price (increase anytime, decrease with notice), Schedule (with notice)

Business Rules

- Price changes: 7-day notice for increases
- Cannot change paid→free if enrollments exist
- Max 5 major edits/month for published courses

- All changes logged (audit trail)
-

UC-008: Delete/Archive Course

Primary Actor: Teacher

Goal: Remove or archive course

Main Flow

1. Teacher selects course → Clicks “Delete” or “Archive”
2. System checks enrolled students → Shows impact summary
3. Teacher confirms → Selects action for students (refund/no refund)
4. System processes refunds (if applicable) → Changes status → Notifies students

Options

- **Delete:** Soft delete (recoverable 30 days), processes refunds
- **Archive:** Hidden from public, students retain access, no refunds

Business Rules

- Cannot delete if upcoming live sessions exist
 - Refund: 0-30% progress = 100%, 31-60% = 50%, 61%+ = No refund
 - Data retained: Student progress 1 year, Content 30 days, Financial records 7 years
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UC-009: Browse Courses

Primary Actor: Student / Guest

Goal: Discover and explore courses

Main Flow

1. User visits courses page
2. System retrieves published courses → Displays grid/list
3. User can:
 - Search by keywords
 - Filter (category, price, level, language, rating, duration)
 - Sort (popular, newest, highest rated, price)
4. User selects course → System shows detail page

Business Rules

- Default: 12 courses/page (grid), 20/page (list)
 - Search: title, description, tags, instructor (case-insensitive, fuzzy matching)
 - Listings cached 5 minutes
 - Guests see all published, logged-in see personalized recommendations
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3. Enrollment & Payment

UC-010: Enroll in Course

Primary Actor: Student

Goal: Register and pay for course

Main Flow

1. Student views course → Clicks “Enroll Now”
2. System verifies login, enrollment status, availability
3. System shows checkout page with price breakdown
4. Student optionally enters discount code → System validates and applies
5. Student reviews refund policy → Agrees to terms → Selects payment method
6. System redirects to payment gateway (Stripe/PayPal)
7. Student completes payment → Gateway processes → Sends webhook
8. System validates webhook → Enrolls student → Grants access → Sends confirmation email

Price Breakdown Example

Course Price:	500 SAR
Discount (20%):	-100 SAR
Subtotal:	400 SAR
VAT (15%):	+ 60 SAR
Total:	460 SAR

Business Rules

- Min price: 10 SAR
 - Platform fee: 10% (from teacher earnings)
 - VAT: 15% (added to student price)
 - Session timeout: 30 minutes
 - Webhook timeout: 10 minutes
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UC-011: Process Payment

Primary Actor: System (Automated)

Goal: Handle payment transaction

Main Flow

1. System receives payment request → Validates → Generates transaction ID
2. Creates payment record (status: Initiated)
3. Calls payment gateway API → Creates payment session
4. Gateway processes transaction → Sends webhook
5. System validates webhook signature → Updates payment record
6. If successful: Enrolls student, sends receipt, logs transaction

Payment Status Flow

Initiated → Processing → Success/Failed/Pending

Business Rules

- Webhook signature mandatory (HMAC-SHA256)
 - Idempotency handling (prevent duplicate processing)
 - Retry: 3 attempts with exponential backoff (1s, 2s, 4s)
 - Transaction logging: 7 years (legal requirement)
 - PCI-DSS Level 1 compliance (via gateways)
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UC-012: Request Refund

Primary Actor: Student / Admin

Goal: Process refund for paid course

Main Flow

1. Student selects enrolled course → Clicks “Request Refund”
2. System checks eligibility → Calculates refund amount
3. Student selects reason → Enters details → Confirms request
4. System creates refund request (status: Pending) → Notifies admin/teacher
5. Admin reviews → Approves or rejects
6. If approved: System initiates refund with gateway → Gateway processes → System updates status
7. System revokes access → Logs transaction → Sends confirmation → Adjusts teacher earnings

Refund Calculation

Days: 0-7 = 100%, 8-14 = 50%, 15+ = No refund

OR

Progress: 0-10% = 100%, 11-30% = 75%, 31-60% = 50%, 61%+ = No refund
(Whichever more favorable to student)

Business Rules

- Auto-approval: <48h enrollment + <10% progress
 - Processing: 5-7 business days
 - Max 3 refunds per student per year
 - Teacher impact: High refund rate (>15%) triggers review
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UC-013: Apply Discount Code

Primary Actor: Student

Goal: Apply discount during checkout

Main Flow

1. Student at checkout → Enters discount code → Clicks “Apply”
2. System validates code (exists, active, not expired, usage limits OK, applicable to course)
3. System calculates discount → Updates price breakdown → Recalculates tax
4. System shows success message → Locks code to transaction

Discount Types

- Percentage (e.g., 20% off)
- Fixed Amount (e.g., 100 SAR off)
- Free Enrollment (100% off)

Business Rules

- Code format: 4-20 alphanumeric chars, case-insensitive
 - Validity: Start/end dates, auto-expires at end date
 - Usage limits: Global limit + per-user limit (typically 1)
 - Cannot combine multiple codes (generally)
 - Teachers create codes for their courses, Admins create platform-wide
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4. Educational Content

UC-014: Add Content to Course

Primary Actor: Teacher

Goal: Add educational materials to course

Main Flow

1. Teacher opens course management → Clicks “Add Content”
2. System shows content type selection
3. Teacher selects type → Fills details → Uploads file/enters content
4. Teacher sets properties (title, description, access type, downloadable, required)
5. System validates → Uploads to cloud → Processes file → Creates record → Links to course

Content Types

1. **Video:** Upload MP4/WebM (max 2GB) or YouTube/Vimeo URL
2. **Document:** Upload PDF (max 50MB), Word, PowerPoint
3. **Quiz:** Create in quiz builder
4. **Text:** Rich text editor with Markdown
5. **External Link:** URL to resource

Business Rules

- Videos: max 2GB/file, 20GB/course, auto-transcode to H.264, multiple qualities (360p, 720p, 1080p)
 - Documents: max 50MB, auto-convert DOCX/PPTX to PDF
 - Storage: Cloudflare R2 or AWS S3 with CDN delivery
 - Free teachers: max 20 lessons/course
 - Backup retention: 30 days after deletion
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UC-015: View Course Content

Primary Actor: Student (Enrolled)

Goal: Access and consume course content

Main Flow

1. Student opens enrolled course → System displays curriculum
2. Student clicks lesson → System checks permissions → Retrieves from CDN
3. System loads appropriate viewer (video player, PDF viewer, text reader)
4. System starts progress tracking → Records viewing time

5. When complete: System updates progress → Unlocks next content (if sequential)

Content Viewers

- **Video:** Custom player with play/pause, speed (0.5x-2x), quality selector, captions, fullscreen
- **PDF:** Embedded viewer with zoom, navigation, search, download (if enabled)
- **Text:** Reading view with font size adjustment, dark mode, read aloud

Business Rules

- Progress tracking: Video 80% = viewed, Document scrolled to end = viewed
 - Sequential access: If enabled, must complete in order
 - Playback speed: Saved per student, applies to all videos
 - Concurrent viewing: One device at a time (5-min grace period for switching)
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UC-016: Track Student Progress

Primary Actor: System (Automated)

Goal: Monitor and record learning progress

Main Flow

1. Student performs activity (view lesson, complete quiz, submit assignment)
2. System detects completion → Records details → Calculates updated progress %
3. System updates database → Triggers unlocks (next lesson, badges) → Updates UI

Tracked Activities

- Lessons viewed, Videos watched (with position), Documents read
- Quizzes attempted/completed, Assignments submitted
- Live class attendance, Time spent, Last access date

Progress Calculation

Overall = (Completed Items / Total Items) × 100%

Weighted =

Videos (40% × completion) +
Quizzes (30% × avg score) +

Assignments (20% × completion) +
Participation (10% × activity)

Business Rules

- Update frequency: Video every 10s, Quiz on submit, Lesson on mark/criteria met
 - Completion criteria: Video 80%, Reading scrolled or time estimated, Quiz submitted
 - Streak tracking: Day counts if 1+ min activity, broken if 24h+ without activity
 - Certificates: Issued at 100% + passing grade + all required assessments
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5. Quizzes & Assessments

UC-017: Create Quiz

Primary Actor: Teacher

Goal: Create assessment for students

Main Flow

1. Teacher navigates to course → Clicks “Add Quiz”
2. System opens quiz builder
3. Teacher enters title/description → Adds questions (select type, enter text, add options, mark correct, set points)
4. Teacher sets quiz settings (time limit, attempts, passing score, show results, randomize)
5. Teacher previews → Saves quiz → System validates → Adds to curriculum

Question Types

1. Multiple Choice (single answer)
2. Multiple Select (multiple correct answers)
3. True/False
4. Fill in the Blank
5. Essay/Long Answer (manual grading)
6. Matching, Ordering

Quiz Settings

- Time Limit: None or 5-120 minutes
- Attempts: 1, 2, 3, or Unlimited
- Passing Score: 0-100% (default: 70%)
- Show Results: Immediately, After due date, Manual release

- Randomize: Question order, Answer options

Business Rules

- Min 1 question, Max 100 questions per quiz, Recommended 10-30
 - Point values: 1-100 per question, total calculated automatically
 - If timed: Enforced strictly, auto-submit when expires, warning at 5 min
 - Objective questions: auto-graded, Essay: manual grading required
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UC-018: Take Quiz

Primary Actor: Student

Goal: Attempt assessment and receive grade

Main Flow

1. Student clicks quiz → System checks eligibility → Shows instructions
2. Student reviews → Clicks “Start Quiz”
3. System generates instance → Randomizes (if set) → Starts timer → Displays questions
4. Student answers questions → System auto-saves every 30s → Navigates between questions
5. Student reviews answers → Clicks “Submit” → Confirms
6. System auto-grades objective → Calculates score → Saves attempt → Updates progress → Shows results (if immediate)

During Quiz

- Question counter, Timer (countdown), Progress indicator
- Navigation (Next, Previous, Review), Flag questions
- Submit button, Auto-save indicator

Results Page

- Score (points/total), Percentage, Pass/Fail
- Correct/Incorrect breakdown, Time taken
- Correct answers (if enabled), Explanations, Feedback

Business Rules

- Navigation: Can move freely, can flag for review, must explicitly submit
- Timing: Grace 5s after expiry, warnings at 10%, 5%, 1%
- Grading: Immediate for objective, delayed for essays
- Scoring: $(\text{Points earned} / \text{Total}) \times 100$, passing grade per quiz, highest score kept

- Cheating prevention: Randomize order, one question at a time (optional), tab-switching detection (optional)
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UC-019: Grade Essay Questions

Primary Actor: Teacher / Assistant

Goal: Manually review and grade essays

Main Flow

1. Student submits quiz with essays → System auto-grades objective → Marks “Pending Manual Grading” → Notifies teacher
2. Teacher opens grading dashboard → Sees pending submissions
3. Teacher selects submission → Reads essay answers → Assigns points → Adds feedback
4. Teacher repeats for all essays → Clicks “Finalize Grade”
5. System calculates final score → Updates grade → Marks “Graded” → Notifies student

Grading Features

- Side-by-side: question and answer
- Rubric display (if created)
- AI-suggested points based on keywords
- Previous answers by same student
- Quick feedback templates, Batch grading

Business Rules

- Grading deadline: Within 7 days, reminder after 3 days, admin notified if >14 days
 - Points: 0 to max, partial credit allowed, bonus points possible
 - Regrading: Student can request within 7 days, only one request per quiz
 - Assistant grading: If authorized, teacher reviews and approves
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6. Live Streaming

UC-020: Start Live Session

Primary Actor: Teacher

Goal: Broadcast live class to students

Main Flow

1. Teacher navigates to scheduled session → Clicks “Start Live Session”
2. System checks time window (± 15 min) → Tests camera/microphone → Shows device selection
3. Teacher selects devices → Tests preview → Clicks “Go Live”
4. System creates streaming room (WebRTC) → Starts streaming → Updates status “Live” → Sends notifications to students

Live Dashboard

- Own video feed, Student count (live), Chat messages
- Raised hands queue, Screen share controls, Recording status
- Audio/video toggles, End session button, Participant list

Streaming Features

- Screen sharing, Whiteboard/drawing tools, Polls and quizzes
- Breakout rooms, Recording (optional), Virtual backgrounds

Business Rules

- Can start 15 min early, must start within 1 hour of scheduled
 - Max duration: 4 hours, warning at 3:45h
 - Max participants: 500 (configurable)
 - Recording: Requires consent, max 10GB/session, stored 1 year
 - Quality: Auto-adjusts (360p, 720p, 1080p), Audio 64-128kbps
-

UC-021: Join Live Session

Primary Actor: Student

Goal: Attend live class

Main Flow

1. Student receives notification or sees “Live” indicator → Clicks “Join Session”
2. System verifies enrollment → Checks device permissions → Shows joining screen
3. Student enables/disables camera & mic → Clicks “Join Now”
4. System connects to streaming room → Loads teacher’s video → Joins voice (muted) → Updates participant list → Records attendance

Student Interface

- Teacher’s video (main), Shared screen (if sharing), Chat panel

- Participant list, Raise hand button, Reactions (, ,)
- Settings (quality), Exit button, Q&A panel

Participation Options

- Watch/listen, Read/send chat, Raise hand, React with emojis
- Answer polls, View shared materials, Take notes

Business Rules

- Must be enrolled, session must be live, not banned
 - Default: Camera off, Microphone muted
 - Attendance: Marked present after 1 min, must stay 75% for “attended”
 - Bandwidth: Min 1 Mbps download, recommended 3+ Mbps
 - Can only join one live session at a time
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UC-022: End Live Session

Primary Actor: Teacher

Goal: Close live session and process post-session tasks

Main Flow

1. Teacher clicks “End Session” → System shows confirmation
2. Teacher confirms → Chooses recording option (save/discard)
3. System stops streaming → Disconnects all students → Stops recording
4. System calculates duration → Processes attendance → Saves chat transcript → Updates status “Ended”
5. If recording saved: System processes video (transcode, thumbnail, timestamps) → Adds to course
6. System generates analytics → Sends summary email → Notifies students

Recording Options

- Save & publish immediately
- Save for review (publish later)
- Keep private (teacher only)
- Discard recording

Session Analytics

- Total duration, Peak/average viewers, Unique participants
- Chat message count, Questions asked, Polls answered
- Engagement score, Attendance report

Business Rules

- Can end anytime (warning if <5 min)
 - Recording: Teacher owns rights, public or private, max 10GB, kept 1 year
 - Attendance: 75% of duration = “attended”, late joins counted from join time
 - Analytics: Available immediately, detailed report within 1 hour
 - Cleanup: Room deleted, temp files cleaned, chat logs saved 90 days
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7. Remaining Sections

Pending Use Cases (54)

Communication (3 UCs)

- UC-023: Send Private Message
- UC-024: Post Question in Course
- UC-025: Send Group Notification

Certificates (2 UCs)

- UC-026: Issue Certificate
- UC-027: Verify Certificate

Reports & Analytics (3 UCs)

- UC-028: View Student Performance Report
- UC-029: View Teacher Analytics
- UC-030: View System Reports (Admin)

AI Chatbot (5 UCs)

- UC-031: Generate Quiz with AI
- UC-032: Create Google Form Automatically
- UC-033: Analyze Student Answers with AI
- UC-034: Get Help from Chatbot (Student)
- UC-035: Generate Educational Content with AI

[... continues with remaining 41 UCs]

Part 3: Appendices

Export Instructions

Quick Export

1. Copy all content (Ctrl+A, Ctrl+C)
2. Paste into Word/Google Docs
3. Apply heading styles
4. Save as DOCX or PDF

Professional Export (Pandoc)

```
pandoc usecases.md -o Athary_UseCases.docx --toc --toc-depth=3
pandoc usecases.md -o Athary_UseCases.pdf --toc --pdf-engine=xelatex
```

Document Status

Progress: 26/80 (32.5%)

Completed: User Management, Course Management, Enrollment & Payment, Educational Content, Quizzes, Live Streaming

Remaining: 54 Use Cases across 23 categories

Estimated completion time: 15-20 hours

Key Features Documented

OAuth 2.0 (Google, Facebook, Apple)
Payment Processing (Stripe, PayPal)
Refund System
Discount Codes
Live Streaming (WebRTC/LiveKit)
Quiz System
Progress Tracking
Cloud Storage (R2/S3)

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Ready for Export and Implementation